

Experiment 8

AIM: Change specification and use any SCM Tool to make different versions

Tools : Git and GitHub

Theory :

Software configuration management: The traditional software configuration management (SCM) process is looked upon by practitioners as the best solution to handling changes in software projects. It identifies the functional and physical attributes of software at various points in time, and performs systematic control of changes to the identified attributes for the purpose of maintaining software integrity and traceability throughout the software development life cycle.

The SCM process further defines the need to trace changes, and the ability to verify that the final delivered software has all of the planned enhancements that are supposed to be included in the release. It identifies four procedures that must be defined for each software project to ensure that a sound SCM process is implemented. They are:

1. Configuration identification
2. Configuration control
3. Configuration status accounting
4. Configuration audits

These terms and definitions change from standard to standard, but are essentially the same.

- Configuration identification is the process of identifying the attributes that define every aspect of a configuration item. A configuration item is a product (hardware and/or software) that has an end-user purpose. These attributes are recorded in configuration documentation and baselined. [Baselining](#) an attribute forces formal configuration change control processes to be effected in the event that these attributes are changed.
- Configuration change control is a set of processes and approval stages required to change a configuration item's attributes and to re-baseline them.
- Configuration status accounting is the ability to record and report on the configuration baselines associated with each configuration item at any moment of time.
- Configuration audits are broken into functional and [physical configuration audits](#). They occur either at delivery or at the moment of effecting the change. A functional configuration audit ensures that functional and performance attributes of a configuration item are achieved, while a physical configuration audit ensures that a configuration item is installed in accordance with the requirements of its detailed design documentation.

GitHub offers all of the distributed revision control and source code management (SCM) functionality of Git as well as adding its own features. Unlike Git, which is strictly a command-line tool, GitHub provides a Web-based graphical interface and desktop as well as mobile integration. It also provides access control and several collaboration features such as bug tracking, feature requests, task management for every project.

SCM For Employee Management System

Figure 1: Git log output showing detailed commit history with timestamps and messages, proving the use of Git for version control and change management.

```
commit cfd36de8f055224c3e0048c45694928c8b87fc84 (HEAD -> master, origin/master, EMS/master)
Author: Unnat <Parvinder221978@gmail.com>
Date: Sat Mar 28 12:41:11 2026 +0530

Transaction changes

commit d6f8be308d97e9a94da5659394a8a91e38f34fdb
Author: Unnat <Parvinder221978@gmail.com>
Date: Fri Feb 6 15:48:29 2026 +0530

Changes done

commit e5f4297d63e74f02f1db760df845877eadc1400e
Author: Unnat <Parvinder221978@gmail.com>
Date: Mon Feb 2 20:56:44 2026 +0530

Changes done

commit be01cdd86559fd5618069b5fb1f6779716a625fe
Author: Unnat <Parvinder221978@gmail.com>
Date: Mon Feb 2 12:18:50 2026 +0530

New changes

commit 218d466f4d34b729660a63f390d303693e70d72a
Author: Unnat <Parvinder221978@gmail.com>
Date: Sun Feb 1 19:30:12 2026 +0530

changes

commit 482f9ff188431203b9ed8e71148860901f3773fc
Author: Unnat <Parvinder221978@gmail.com>
Date: Sat Jan 31 12:45:09 2026 +0530

More changes

commit 7a1f27bf7907782043dcde4e0674ab8004e357b9
Author: Unnat <Parvinder221978@gmail.com>
Date: Thu Jan 29 20:58:07 2026 +0530

Mail changes and Cn changes

commit a5fe8f68a3406ece809792332b31d921fdf87153
Author: Unnat <Parvinder221978@gmail.com>
Date: Wed Jan 28 18:32:06 2026 +0530

Pending action and mail
```



TCET
DEPARTMENT OF COMPUTER ENGINEERING
Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education
Under Autonomy - CBCGS-HME 2023
University of Mumbai



Figure 2: GitHub repository showing project structure, files, and total number of commits, demonstrating successful project setup using GitHub.

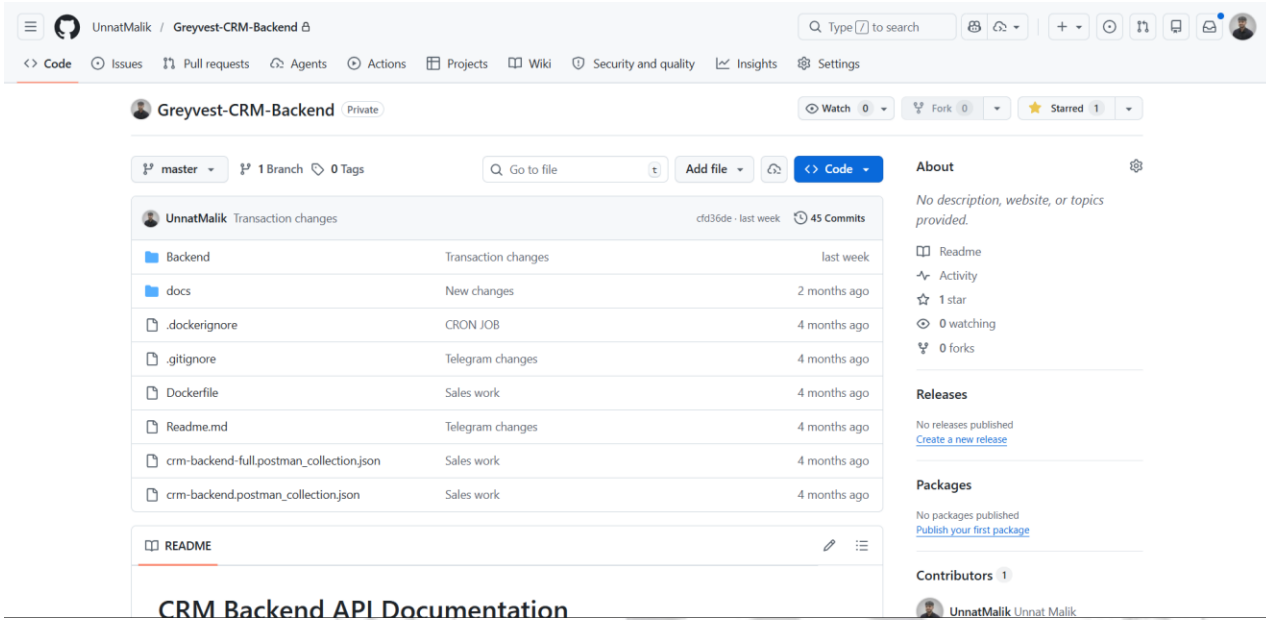


Figure 3: Commit history displaying multiple commits with different messages, indicating continuous updates and version tracking of the project.

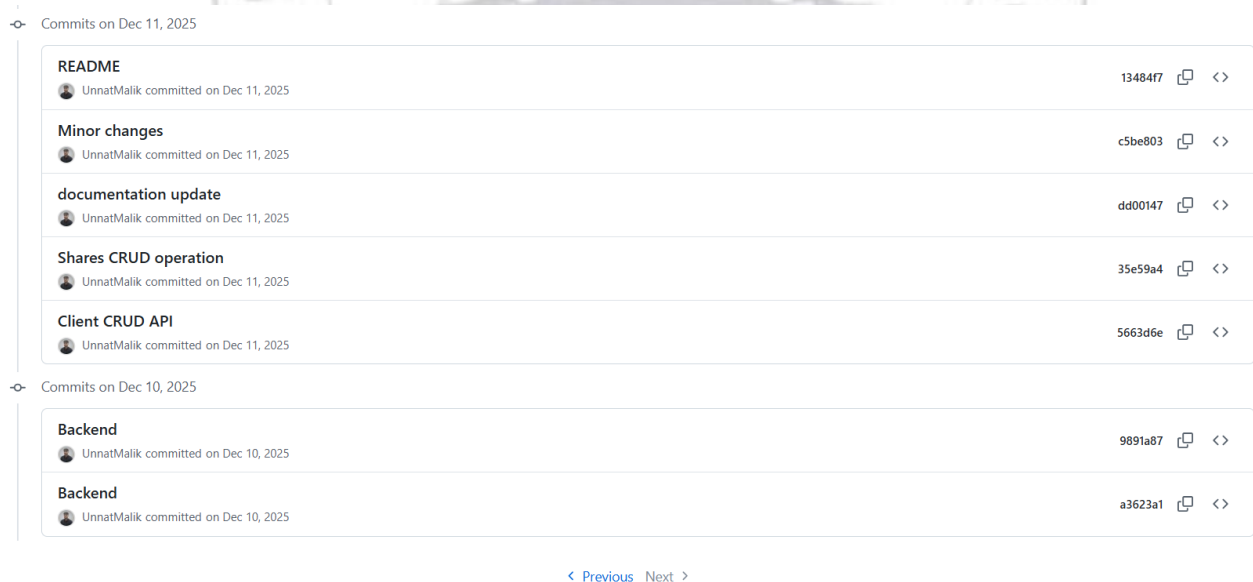




Figure 4: Commit timeline showing changes made on different dates with feature updates, demonstrating modification of project specifications over time.

Commits on Jan 20, 2026	Buy order model and mail creation UnnatMalik committed on Jan 20 ea6cf1d
Commits on Jan 19, 2026	Mail integration UnnatMalik committed on Jan 19 5fb9836
Commits on Jan 17, 2026	Changes UnnatMalik committed on Jan 17 bb6f075
Commits on Jan 14, 2026	Changes in perfroma and creating endpoints for calling for admin UnnatMalik committed on Jan 14 20f9dcb Changes in perfroma and creating endpoints for calling for admin UnnatMalik committed on Jan 14 f1e3d9d
Commits on Jan 13, 2026	New API's UnnatMalik committed on Jan 13 e56e003
Commits on Jan 2, 2026	Sales Logic almost commit UnnatMalik committed on Jan 2 261fa83



Learning Outcomes: Students should have the ability to

LO1: Understand the concept of Software Configuration Management (SCM) and its importance in software development.

LO2: Analyze and implement version control using tools like Git and GitHub.

LO3: Apply techniques such as committing changes, maintaining version history, and tracking modifications in a project.

LO4: Demonstrate the use of branching and commit logs to manage different versions of software efficiently.

Outcomes: Upon completion of this experiment, students will be able to develop and manage multiple versions of a project using SCM tools. They will be capable of tracking and recording changes in project specifications through commits, maintaining proper version history, and ensuring traceability of software changes. Furthermore, students will be able to explain the working of version control systems with proper documentation and workflow representation.

Conclusion:

For Faculty Use

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	
Marks Obtained				